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COMMUNICATION DEVICE AND COMMUNICATION SYSTEM

Abstract

A communication device according to one embodiment of the present disclosure includes: a detector configured to detect a signal input that comes from outside; and a changer configured to use a fact that there is a detection in the detector as a trigger to change a Connection Interval for Bluetooth (registered trademark) Low Energy.

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Background/Summary

TECHNICAL FIELD
[0001] The present disclosure relates to a communication device and a communication system.
BACKGROUND ART
[0002] As a communication standard for performing wireless communication with extremely low consumption of electric power, Bluetooth (registered trademark) Low Energy (hereinafter referred to as BLE) is known. For example, Patent Literature 1 describes wireless communication by using BLE.
CITATION LIST
Patent Literature
[0003] Patent Literature 1: Japanese Unexamined Patent Application Publication No. 2021-61460
SUMMARY OF THE INVENTION
[0004] Incidentally, there is an issue that wireless communication by using BLE, due to its lower responsiveness, is not suitable for applications where higher responsiveness is demanded. Therefore, it is desirable to provide a communication device and a communication system that make it possible to acquire responsiveness in accordance with an application.
[0005] A communication device according to one embodiment of the present disclosure includes: a detector configured to detect a signal input that comes from outside; and a changer configured to use a fact that there is a detection in the detector as a trigger to change a Connection Interval for Bluetooth (registered trademark) Low Energy.
[0006] A communication system according to one embodiment of the present disclosure includes a master device and a slave device communicable to each other by using a Bluetooth (registered trademark) Low Energy communication. The master device includes: a detector configured to detect a signal input that comes from outside; and a communicator configured to use a fact that there is a detection in the detector as a trigger to transmit, to the slave device by using a BLE communication, a change instruction for a value of a Connection Interval for BLE.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0007] FIG. **1** is a diagram illustrating an example of an outline configuration of a communication system according to a first embodiment of the present disclosure.
[0008] FIG. **2** is a diagram illustrating an example of functional blocks of a terminal device illustrated in FIG. **1**.
[0009] FIG. **3** is a diagram illustrating an example of functional blocks of a left-side earphone illustrated in FIG. **1**.
[0010] FIG. **4** is a diagram illustrating an example of functional blocks of a right-side earphone illustrated in FIG. **1**.
[0011] FIG. **5** is a diagram illustrating an example of a communication procedure for the left-side earphone and the right-side earphone in the communication system illustrated in FIG. **1**.
[0012] FIG. **6** is a diagram illustrating an example of a communication procedure for the left-side earphone and the right-side earphone in the communication system illustrated in FIG. **1**.
[0013] FIG. **7** is a diagram illustrating an example of a communication procedure for the left-side earphone and the right-side earphone in the communication system illustrated in FIG. **1**.
[0014] FIG. **8** is a diagram illustrating an example of a communication procedure for the left-side earphone and the terminal device in the communication system illustrated in FIG. **1**.
[0015] FIG. **9** is a diagram illustrating an example of an outline configuration of a communication

system according to a second embodiment of the present disclosure.

[0016] FIG. **10** is a diagram illustrating an example of functional blocks of a smart speaker illustrated in FIG. **9**.

[0017] FIG. **11** is a diagram illustrating an example of a communication procedure for the smart speaker and the terminal device in the communication system illustrated in FIG. **9**.

MODES FOR CARRYING OUT THE INVENTION

[0018] Hereinafter, embodiments for practicing the present disclosure are described in detail with reference to the drawings.

1. Background of BLE

[0019] When a central device performs coupling with respect to a peripheral device in a state where the peripheral device is outputting a beacon (advertise), the central serves as a master, and the peripheral serves as a slave. At this time, there is a transition in communication face from an advertise channel to a data channel. As there is a transition in communication face to the data channel, the master outputs data packets at constant intervals to the slave at a master's timing. The timing is called a Connection Event. The interval at which data packets are to be outputted is called a Connection Interval. It is possible to set a value of the Connection Interval within a range between 7.5 ms to 4 sec. The Connection Interval is shared between the master and the slave.

[0020] In a case where a Connection Interval is shorter, responsiveness and a communication speed increase, but consumption of electric power also increases. In a case where a Connection Interval is longer, responsiveness and a communication speed decrease, but consumption of electric power is suppressed to low. Therefore, it is difficult to use a communication device, in a case where such a value that extends a Connection Interval longer is set, for an application where higher responsiveness is demanded. Then, the inventors of the present application propose a communication device and a communication system including the communication device that make it possible to acquire responsiveness in accordance with an application, which are described below.

2. First Embodiment

[Configuration]

[0021] A communication system **1** according to a first embodiment of the present disclosure will now be described herein. FIG. **1** illustrates an outline configuration example of the communication system **1**. The communication system **1** includes, for example, as illustrated in FIG. **1**, a wireless earphone device **100** and a terminal device **200**. The wireless earphone device **100** includes, for example, as illustrated in FIG. **1**, a left-side earphone **110** and a right-side earphone **120**. The left-side earphone **110** corresponds to one specific example of a “master device” according to the present disclosure. The right-side earphone **120** corresponds to one specific example of a “slave device” according to the present disclosure.

[0022] The left-side earphone **110**, the right-side earphone **120**, and the terminal device **200** are communicable to each other by using a BLE standard that is one of extended specifications relating to Bluetooth (registered trademark) that is one of wireless communication standards stipulated by a special interest group (SIG). In FIG. **1**, a communication network based on BLE, which is formed between the left-side earphone **110** and the right-side earphone **120** and the terminal device **200**, is expressed as BLE**1**, and a communication network based on BLE, which is formed between the left-side earphone **110** and the right-side earphone **120**, is expressed as BLE**2**.

[0023] The terminal device **200** is configured to transmit data including music data, for example, with respect to the wireless earphone device **100** via the communication network BLE**1**. The terminal device **200** includes, for example, as illustrated in FIG. **2**, a user inputter **201**, a storage **202**, a display **203**, an arithmetic processor **204**, and a communicator **205**.

[0024] The user inputter **201** includes, for example, an input interface including a touch panel. The storage **202** includes, for example, a volatile memory such as a dynamic random-access memory (DRAM) or a non-volatile memory such as an electrically erasable programmable read-only memory (EEPROM) or a flash memory. The storage **202** stores, for example, a program describing a series of processing for operating the wireless earphone device **100** in a remote manner. The arithmetic processor **204** is configured to execute the program read from the storage **202**, for example, to execute the series of processing for operating the wireless earphone device **100** in a remote manner. The display **203** is configured to display, for example, content for supporting an input to be performed into the user inputter **201**. The communicator **205** includes an interface configured to perform communication with the wireless earphone device **100** via the communication network BLE**1**. The communicator **205** is configured to transmit data including music data, for example, with respect to the wireless earphone device **100** via the communication network BLE**1**. The communicator **205** is configured to receive data including a command, for example, from the wireless earphone device **100** via the communication network BLE**1**.

(Left-Side Earphone **110**)

[0025] The left-side earphone **110** is attached to a left ear of a person, and includes, for example, as illustrated in FIG. **3**, a vibration sensor **111**, a peak detector **112**, a time measurer **113**, a communicator **114**, an arithmetic processor **115**, a communicator **116**, a sound processor **117**, and a speaker **118**. The vibration sensor **111** corresponds to one specific example of a “detector configured to detect a signal input that comes from outside” and a “first detector configured to detect a signal input that comes from outside” according to the present disclosure. The communicator **114** corresponds to one specific example of a “first communicator” according to the present disclosure.

[0026] The vibration sensor **111** is configured to detect vibrations of the left-side earphone **110**, and output a vibration signal acquired through the detection to the peak detector **112**. The vibration sensor **111** includes, for example, a gyro sensor configured to detect an angular velocity of the left-side earphone **110**. The vibration sensor **111** is, for example, configured to generate a vibration signal on the basis of a signal of the angular velocity detected by the gyro sensor.

[0027] The peak detector **112** is configured to detect a peak included in the vibration signal inputted from the vibration sensor **111**. The peak detector **112** is configured to output a peak signal acquired through the peak detection to the time measurer **113** and the communicator **114**. The peak detector **112** is further configured to output peak information acquired through the peak detection to the communicator **114**. The peak information includes at least a magnitude of the peak (a vibration amplitude value).

[0028] The peak detector **112** includes, for example, a comparator. The comparator is configured to generate a digital signal of “1” when a value of a vibration signal exceeds a predetermined threshold value, and generate a digital signal of “0” when a value of a vibration signal is equal to or less than the predetermined threshold value. The peak detector **112** is, for example, configured to output the digital signal of “1” generated by the comparator as a peak signal. The peak detector **112** is, for example, further configured to detect a maximum value of the vibration signal within a predetermined period of time from when the peak signal is outputted from the comparator, and output the detected maximum value as a vibration amplitude value.

[0029] The time measurer **113** is configured to use an input of the peak signal from the peak detector **112** as a trigger to start time measurement. The time measurer **113** is configured to output a measurement start time to the communicator **114**. The time measurer **113** is further configured to output a change end signal to the communicator **114** in a case where there is no input of a new peak signal until a predetermined period of time has passed from the start of the measurement. The change end signal is a signal indicating that a changed value of a Connection Interval is to be returned to an original value. The “predetermined period of time” for outputting the change end signal is set in advance by a user, for example. The time measurer **113** includes, for example, a timer configured to output pulses at predetermined cycles and a counter configured to count the pulses outputted from the timer. The time measurer **113** is, for example, configured to use an input of the peak signal as a trigger, cause the counter to count the pulses outputted from the timer, and output the change end signal when a value of the count exceeds a predetermined threshold value corresponding to the “predetermined period of time”.

[0030] The communicator **114** is configured to perform communication with the right-side earphone **120** (specifically, a communicator **124** described later) via the communication network BLE**2**. The communicator **114** is configured to use an input of the peak signal from the peak detector **112** as a trigger to execute the processing for changing the Connection Interval for BLE. That is, the communicator **114** is configured to serve as a master in BLE communication with the right-side earphone **120** via the communication network BLE**2**. The communicator **114** includes, for example, a communication interface circuit conforming to BLE, which is configured to execute the processing. The processing in the communication interface circuit will be described later in detail.

[0031] The communicator **114** is further configured to output first vibration detection information inputted from the peak detector **112** and the time measurer **113** and second vibration detection information inputted from the right-side earphone **120** to the arithmetic processor **115**. The first vibration detection information includes the peak information (the vibration amplitude value) inputted from the peak detector **112** and the measurement start time inputted from the time measurer **113**. The second vibration detection information includes peak information (a vibration amplitude value) acquired by a peak detector **122** described later and a measurement start time acquired by a time measurer **123** described later.

[0032] The arithmetic processor **115** is configured to execute command generation processing on the basis of the first and second vibration detection information inputted from the communicator **114**. The arithmetic processor **115** is, for example, configured to determine whether or not there is a command that has been erroneously detected on the basis of the first and second vibration detection information inputted from the communicator **114**, and execute the command generation processing in accordance with a result of the determination. A method for the determination and the command generation processing will be described later in detail. The arithmetic processor **115** may be configured to output predetermined sound data to the sound processor **117** for the command generation processing. The arithmetic processor **115** may be, for example, configured to output electronic sound data indicating reception of a command and a type of the received command to the sound processor **117**. The arithmetic processor **115** is further configured to output, in a case where a command for providing a request to the terminal device **200** is included as a result of the command generation processing, data including the command, for example, to the communicator **116**.

[0033] The communicator **116** is configured to perform communication with the terminal device **200** (the communicator **205**) via the communication network BLE**1**. The communicator **116** is, for example, configured to receive music data from the terminal device **200** (the communicator **205**) via the communication network BLE**1**. The communicator **116** is, for example, configured to output the music data received from the terminal device **200** (the communicator **205**) to the sound processor **117**. The communicator **116** is configured to transmit the data including the command, for example, which is received from the arithmetic processor **115**, to the terminal device **200** (the communicator **205**) via the communication network BLE**1**.

[0034] The sound processor **117** is, for example, configured to convert the electronic sound data inputted from the arithmetic processor **115** and the music data inputted from the communicator **116** into drive signals, and output the converted drive signals to the speaker **118**. The speaker **118** is configured to output sound on the basis of the drive signals inputted from the sound processor **117**.

(Right-Side Earphone **120**)

[0035] The right-side earphone **120** is attached to a right ear of the person, and includes, for example, as illustrated in FIG. **4**, a vibration sensor **121**, the peak detector **122**, the time measurer **123**, the communicator **124**, an arithmetic processor **125**, a communicator **126**, a sound processor **127**, and a speaker **128**. The vibration sensor **121** corresponds to one specific example of a “second detector configured to detect a signal input that comes from outside” according to the present

disclosure. The communicator **124** corresponds to one specific example of a "second communicator" according to the present disclosure.
[0036] The vibration sensor **121** is configured to detect vibrations of the right-side earphone **120**, and output a vibration signal acquired through the detection to the peak detector **122**. The vibration sensor **121** includes, for example, a gyro sensor configured to detect an angular velocity of the right-side earphone **120**. The vibration sensor **121** is, for example, configured to generate a vibration signal on the basis of a signal of the angular velocity detected by the gyro sensor.
[0037] The peak detector **122** is configured to detect a peak included in the vibration signal inputted from the vibration sensor **121**. The peak detector **122** is configured to output a peak signal acquired through the peak detection to the time measurer **123**. The peak detector **122** is further configured to output peak information acquired through the peak detection to the communicator **124**. The peak information includes at least a magnitude of the peak (a vibration amplitude value).
[0038] The peak detector **122** includes, for example, a comparator. The comparator is configured to generate a digital signal of "1" when a value of a vibration signal exceeds a predetermined threshold value, and generate a digital signal of "0" when a value of a vibration signal is equal to or less than the predetermined threshold value. The peak detector **122** is, for example, configured to output the digital signal of "1" generated by the comparator as a peak signal. The peak detector **122** is, for example, further configured to detect a maximum value of the vibration signal within a predetermined period of time from when the peak signal is outputted from the comparator, and output the detected maximum value as a vibration amplitude value.
[0039] The time measurer **123** is configured to use an input of the peak signal from the peak detector **122** as a trigger to start time measurement. The time measurer **123** is configured to output a measurement start time to the communicator **124**. The time measurer **123** is further configured to output a change end signal to the communicator **114** in a case where there is no input of a new peak signal until a predetermined period of time has passed from the start of the measurement. The "predetermined period of time" for outputting the change end signal is set in advance by the user, for example. The time measurer **123** includes, for example, a timer configured to output pulses at predetermined cycles and a counter configured to count the pulses outputted from the timer. The time measurer **123** is, for example, configured to use an input of the peak signal as a trigger, cause the counter to count the pulses outputted from the timer, and output the change end signal when a value of the count exceeds a predetermined threshold value corresponding to the "predetermined period of time".
[0040] The communicator **124** is configured to perform communication with the left-side earphone **110** (the communicator **114**) via the communication network BLE**2**. The communicator **124** is configured to execute the processing for changing the Connection Interval for BLE when a Connection Interval change instruction for BLE is received from the left-side earphone **110** via the communication network BLE**2**. That is, communicator **124** is configured to serve as a slave in BLE communication with the left-side earphone **110** via the communication network BLE**2**. The communicator **114** includes, for example, a communication interface circuit conforming to BLE, which is configured to execute the processing. The processing in the communication interface circuit will be described later in detail.
[0041] The communicator **124** is further configured to transmit the second vibration detection information inputted from the peak detector **122** and the time measurer **123** to the left-side earphone **110** via the communication network BLE**2** in response to a request provided from the left-side earphone **110**. The second vibration detection information includes, as described above, the peak information (the vibration amplitude value) acquired by the peak detector **122** and the measurement start time acquired by the time measurer **123**.
[0042] The arithmetic processor **125** is, for example, configured to output predetermined data to the communicator **126** and the sound processor **127**, as necessary.
[0043] The communicator **126** is configured to perform communication with the terminal device

200 (the communicator **205**) via the communication network BLE**1**. The communicator **126** is, for example, configured to receive music data from the terminal device **200** (the communicator **205**) via the communication network BLE**1**. The communicator **126** is, for example, configured to output the music data received from the terminal device **200** (the communicator **205**) to the sound processor **127**. The communicator **126** is, for example, configured to transmit the data received from the arithmetic processor **115** to the terminal device **200** (the communicator **205**) via the communication network BLE**1**, as necessary.

[0044] The sound processor **127** is, for example, configured to convert the data inputted from the arithmetic processor **125** and the music data inputted from the communicator **126** into drive signals, and output the converted drive signals to the speaker **128**. The speaker **128** is configured to output sound on the basis of the drive signals inputted from the sound processor **127**.

[Operation]

[0045] Next, operation in the communication system **1** will now be described herein.

[0046] FIG. **5** illustrates an example of a communication procedure for the left-side earphone **110** (the master) and the right-side earphone **120** (the slave) in the communication system **1**. Note that it is assumed that a data channel serves as a communication face.

[0047] It is first assumed that the user has tapped his or her left cheek (the cheek on the left-side earphone **110** side). Vibrations on the cheek, which have occurred due to the tapping, are then transmitted to the left-side earphone **110**, and the vibration sensor **111** detects the vibrations, and outputs a vibration signal to the peak detector **112** (step S**101**). The peak detector **112** detects a peak included in the vibration signal inputted from the vibration sensor **111**, and outputs a peak signal to the communicator **114**. The left-side earphone **110** (the communicator **114**) uses an input of the peak signal from the peak detector **112** as a trigger, and changes the Connection Interval for BLE. Then, the left-side earphone **110** (the communicator **114**) uses an input of the peak signal from the peak detector **112** as a trigger to transmit a change instruction for the Connection Interval for BLE to, and the left-side earphone **110** (the communicator **114**) uses an input of the peak signal from the peak detector **112** as a trigger to transmit a change instruction for the Connection Interval for BLE to the right-side earphone **120** (the communicator **124**) via the communication network BLE**2** (step S**102**).

[0048] At this time, the change instruction is, for example, LL_CONNECTION_UPDATE_IND (a command for Link Layer). The command includes at least a value of the Connection Interval (a value within a range between 7.5 ms to 4 sec).

[0049] The right-side earphone **120** (the communicator **124**) receives the change instruction for the Connection Interval for BLE from the left-side earphone **110** via the communication network BLE**2** (step S**103**). Then, the communicator **124** changes, after waiting for a period of time determined in the BLE standard (6 Connection Events), the value of the Connection Interval for BLE to the value included in the change instruction (steps S**104**, S**105**).

[0050] After the series of processing has been completed, BLE communication is performed between the left-side earphone **110** (the communicator **114**) and the right-side earphone **120** (the communicator **124**) at the Connection Interval after changed.

[0051] In the communication procedure described above, the master has taken an initiative to perform communication. However, it is possible to use a request provided from the slave as a trigger to perform communication, as described below.

[0052] FIG. **6** illustrates an example of a communication procedure for the left-side earphone **110** (the master) and the right-side earphone **120** (the slave) in the communication system **1**. Note that it is assumed that a data channel serves as a communication face.

[0053] It is first assumed that the user has tapped his or her right cheek (the cheek on the right-side earphone **120** side). Vibrations on the cheek, which have occurred due to the tapping, are then transmitted to the right-side earphone **120**, and the vibration sensor **121** detects the vibrations, and outputs a vibration signal to the peak detector **122** (step S**201**). The peak detector **122** detects a peak included in the vibration signal inputted from the vibration sensor **121**, and outputs a peak signal to the communicator **124**. The right-side earphone **120** (the communicator **124**) uses an input of the peak signal from the peak detector **122** as a trigger to transmit a Connection Parameter change request for BLE to the left-side earphone **110** (the communicator **114**) via the communication network BLE**2** (step S**202**). At this time, the change request is, for example, L2CAP_CONNECTION PARAMETER UPDATE REQUEST.

[0054] The left-side earphone **110** (the communicator **114**) receives the Connection Parameter change request for BLE from the right-side earphone **120** via the communication network BLE**2** (step S**203**). Then, the left-side earphone **110** (the communicator **114**) transmits a ConnectionParameter change response for BLE to the right-side earphone **120** (the communicator **124**) via the communication network BLE**2** (step S**204**). At this time, the change response is, for example, L2CAP_CONNECTION PARAMETER UPDATE RESPONSE (ACCEPT).

[0055] The right-side earphone **120** (the communicator **124**) receives the ConnectionParameter change response for BLE from the left-side earphone **110** (the communicator **114**) via the communication network BLE**2** (step S**205**). Thus, an environment for transmitting a Connection Interval change instruction for BLE is prepared.

[0056] After that, the left-side earphone **110** (the communicator **114**) changes the Connection Interval for BLE. The left-side earphone **110** (the communicator **114**) further transmits a change instruction for the Connection Interval for BLE to the right-side earphone **120** (the communicator **124**) via the communication network BLE**2** (step S**206**).

[0057] At this time, the change instruction is, for example, LL_CONNECTION_UPDATE_IND (a command for Link Layer). The command includes at least a value of the Connection Interval (a value within a range between 7.5 ms to 4 sec).

[0058] The right-side earphone **120** (the communicator **124**) receives the change instruction for the Connection Interval for BLE from the left-side earphone **110** via the communication network BLE**2** (step S**207**). Then, the communicator **124** changes, after waiting for a period of time determined in the BLE standard (6 Connection Events), the value of the Connection Interval for BLE to the value included in the change instruction (steps S**208**, S**209**).

[0059] After the series of processing has been completed, BLE communication is performed between the left-side earphone **110** (the communicator **114**) and the right-side earphone **120** (the communicator **124**) at the Connection Interval after changed.

[0060] Next, an example of BLE communication at the Connection Interval after changed will now be described herein. In a case where the value of the Connection Interval has been changed to a smaller value after having undergone the procedure described above, it is possible to execute processing where higher responsiveness is demanded. Examples of the processing where higher responsiveness is demanded include a determination of whether or not there is a command that has been erroneously detected and the command generation processing in accordance with a result of the determination, which are described above.

[0061] FIG. **7** illustrates an example of a communication procedure for the left-side earphone **110** and the right-side earphone **120** in the communication system **1**. Note that it is assumed that a data channel serves as a communication face.

[0062] It is first assumed that the user has tapped his or her left cheek (the cheek on the left-side earphone **110** side). Vibrations on the cheek, which have occurred due to the tapping, are then transmitted to the left-side earphone **110**, and the vibration sensor **111** detects the vibrations, and outputs a vibration signal to the peak detector **112** (step S**301**). Next, the left-side earphone **110** and the right-side earphone **120** execute steps S**102** to S**105** in the processing (Connection Interval change processing) (step S**302**).

[0063] Next, the left-side earphone **110** (the communicator **114**) transmits a vibration detection information transmission request to the right-side earphone **120** (the communicator **124**) via the communication network BLE**2** (step S**303**). As the vibration detection information transmission request is received from the left-side earphone **110** (the communicator **114**) via the communication network BLE**2** (step S**304**), the right-side earphone **120** (the communicator **124**) transmits second vibration detection information to the left-side earphone **110** (the communicator **114**) via the communication network BLE**2** (step S**305**). The left-side earphone **110** (the communicator **114**) receives the second vibration detection information as a response to the vibration detection information transmission request from the left-side earphone **110** (the communicator **114**) via the communication network BLE**2** (step S**306**). Then, the arithmetic processor **115** determines whether or not there is a command that has been erroneously detected on the basis of the received second vibration detection information and the first vibration detection information, and executes the command generation processing in accordance with a result of the determination (step S**307**).

[0064] The arithmetic processor **115**, for example, compares the measurement start time acquired by the time measurer **113** and the measurement start time acquired by the time measurer **123** with each other, and determines which one of the measurement start times corresponds to an earlier time. The arithmetic processor **115**, for example, further compares the peak information (the vibration amplitude value) acquired by the peak detector **112** and the peak information (the vibration amplitude value) acquired by the peak detector **122** with each other, and determines which one of the pieces of peak information (the vibration amplitude values) has a greater value. It is assumed, as a result, that the measurement start time acquired by the time measurer **113** be earlier than the measurement start time acquired by the time measurer **123**, and the peak information (the vibration amplitude value) acquired by the peak detector **112** be greater than the peak information (the vibration amplitude value) acquired by the peak detector **122**. In this case, the arithmetic processor **115**, for example, determines that the first vibration detection information is true, and the second vibration detection information is regarded as information that the user does not desire, and utilizes the first vibration detection information in later steps of the command generation processing.

[0065] FIG. **8** illustrates an example of a communication procedure for the left-side earphone **110** and the terminal device **200** in the communication system **1**. Note that it is assumed that a data channel serves as a communication face. Note that the terminal device **200** serves as a master in BLE communication with the left-side earphone **110** via the communication network BLE**1**. The left-side earphone **110** serves as a slave in BLE communication with the terminal device **200** via the communication network BLE**1**.

[0066] The left-side earphone **110** and the terminal device **200** prepare an environment for transmitting a command on the basis of a request provided from the slave side. After that, the left-side earphone **110** transmits the command generated through the command generation processing described above to the terminal device **200** via the communication network BLE**1** (step S**401**). The terminal device **200** receives the command from the left-side earphone **110** via the communication network BLE**1** (step S**402**), and executes the received command (step S**403**).

[0067] Note that, in a case where the time measurer **113** has outputted a change end signal to the communicator **114**, the left-side earphone **110** (the communicator **114**) uses an input of the change end signal as a trigger to change the value of the Connection Interval for BLE to the value of the Connection Interval before the change. The left-side earphone **110** (the communicator **114**) further uses the input of the change end signal as a trigger to transmit a change instruction for the Connection Interval for BLE to the right-side earphone **120** (the communicator **124**) via the communication network BLE**2**. At this time, the change instruction includes the value of the Connection Interval before the change.

[0068] The right-side earphone **120** (the communicator **124**) receives the change instruction for the Connection Interval for BLE from the left-side earphone **110** via the communication network BLE**2**. Then, the communicator **124** changes, after waiting for a period of time determined in the BLE standard (6 Connection Events), the value of the Connection Interval for BLE to the value included in the change instruction.

[0069] After the series of processing has been completed, BLE communication is performed between the left-side earphone **110** (the communicator **114**) and the right-side earphone **120** (the communicator **124**) at the Connection Interval before the change.

[Effects]

[0070] Next, effects of the communication system **1** will now be described herein.

[0071] In the present embodiment, a fact that there is a detection of vibrations on at least either one of the left-side earphone **110** and the right-side earphone **120** (a peak detection) is used as a trigger to change the Connection Interval for BLE. Thus, it is possible to perform communication between the left-side earphone **110** and the right-side earphone **120** on the basis of a detection of vibrations at higher responsiveness. Therefore, it is possible to acquire responsiveness in accordance with an application.

[0072] In the present embodiment, a value of the Connection Interval, which is changed by the left-side earphone **110** (the communicator **114**), is transmitted to the right-side earphone **120** (the communicator **124**) via the communication network BLE**2**. Thus, it is possible to perform communication between the left-side earphone **110** and the right-side earphone **120** on the basis of a detection of vibrations at higher responsiveness. Therefore, it is possible to acquire responsiveness in accordance with an application.

[0073] In the present embodiment, in a case where there is no detection of vibrations on at least either one of the left-side earphone **110** and the right-side earphone **120** (a peak detection) for a predetermined period of time, the value of the Connection Interval is returned to the value before the change. Thus, it is possible to suppress excessive consumption of electric power.

3. Second Embodiment

[0074] Next, a communication system **2** according to a second embodiment of the present disclosure will now be described herein. FIG. **9** illustrates an outline configuration example of the communication system **2**. The communication system **2** includes, for example, as illustrated in FIG. **9**, a smart speaker **300** and the terminal device **200**. The smart speaker **300** corresponds to one specific example of a “master device” according to the present disclosure. The terminal device **200** corresponds to one specific example of a “slave device” according to the present disclosure. The smart speaker **300** and the terminal device **200** are communicable to each other by using a BLE standard. In FIG. **9**, a communication network based on BLE, which is formed between the smart speaker **300** and the terminal device **200**, is expressed as BLE**3**.

[0075] The terminal device **200** is configured to transmit data including music data, for example, with respect to the smart speaker **300** via the communication network BLE**3**. The smart speaker **300** is a device to which it is possible to input sound, and includes, for example, as illustrated in FIG. **10**, a microphone **301**, an utterance processor **302**, a time measurer **303**, a communicator **304**, an arithmetic processor **305**, a sound processor **306**, and a speaker **307**. The microphone **301** corresponds to one specific example of a “detector configured to detect a signal input that comes from outside” according to the present disclosure.

[0076] The microphone **301** is a device configured to detect ambient sound (for example, an utterance that has occurred outside), and is configured to output a sound signal acquired through the detection to the utterance processor **302**.

[0077] The utterance processor **302** is configured to analyze the sound signal inputted from the microphone **301**, and determine whether or not the sound signal includes a signal component corresponding to a predetermined sound command. Examples of the “predetermined sound command” include a start command that means starting of the smart speaker **300** and a function command for executing a predetermined function of the terminal device **200**. Examples of the function command include a music playing command that instructs playing of designated music data.

[0078] The utterance processor **302** is configured to output, in a case where the sound signal includes a signal component corresponding to the start command, as a result of the determination, a start signal meaning that the start command has been inputted to the time measurer **303** and the communicator **304**. The utterance processor **302** is further configured to output the start command to the communicator **304**. The utterance processor **302** is configured to output, in a case where the sound signal includes a signal component corresponding to a function command, as a result of the determination, a function signal meaning that the function command has been inputted to the time measurer **303** and the communicator **304**. The utterance processor **302** is further configured to output the function command to the communicator **304**.

[0079] The time measurer **303** is configured to use an input of the start signal from the utterance processor **302** as a trigger to start time measurement. The time measurer **303** is configured to output a change end signal to the communicator **304** in a case where there is no input of a new signal (for example, the function signal) until a predetermined period of time has passed from the start of the measurement. The change end signal is a signal indicating that a changed value of a Connection Interval is to be returned to an original value. The "predetermined period of time" for outputting the change end signal is set in advance by the user, for example. The time measurer **303** includes, for example, a timer configured to output pulses at predetermined cycles and a counter configured to count the pulses outputted from the timer. The time measurer **303** is, for example, configured to use an input of the start signal as a trigger, cause the counter to count the pulses outputted from the timer, and output the change end signal when a value of the count exceeds a predetermined threshold value corresponding to the "predetermined period of time".

[0080] The communicator **304** is configured to perform communication with the terminal device **200** (the communicator **205**) via the communication network BLE**3**. The communicator **304** is configured to use an input of the start signal from the utterance processor **302** as a trigger to execute the processing for changing the Connection Interval for BLE. That is, the communicator **304** is configured to serve as a master in BLE communication with the terminal device **200** via the communication network BLE**3**. The communicator **304** includes, for example, a communication interface circuit conforming to BLE, which is configured to execute the processing. The processing in the communication interface circuit will be described later in detail.

[0081] The communicator **304** is configured to transmit, when the function command is inputted from the utterance processor **302**, the inputted function command to the terminal device **200** (the communicator **205**) via the communication network BLE**3**. The communicator **304** is configured to receive, in a case where the function command transmitted to the terminal device **200** (the communicator **205**) is the music playing command, for example, music streaming data from the terminal device **200** (the communicator **205**). The communicator **304** is configured to at this time output the received music streaming data to the arithmetic processor **305**.

[0082] The arithmetic processor **305** is, for example, configured to output predetermined data to the communicator **304** and the sound processor **306**, as necessary. The arithmetic processor **305** is, for example, configured to output the music streaming data received from the communicator **304** to the sound processor **306**.

[0083] The sound processor **306** is, for example, configured to convert the data inputted from the arithmetic processor **305** into drive signals, and output the converted drive signals to the speaker **307**. The speaker **307** is configured to output sound on the basis of the drive signals inputted from the sound processor **306**.

[Operation]

[0084] Next, operation in the communication system **2** will now be described herein.

[0085] FIG. **11** illustrates an example of a communication procedure for the smart speaker **300** (the master) and the terminal device **200** (the slave) in the communication system **2**. Note that it is assumed that a data channel serves as a communication face.

[0086] It is first assumed that the user has uttered the start command toward the smart speaker **300**. Then, the microphone **301** detects the utterance of the user, and the utterance processor **302** detects the start command from a sound signal (step S**501**). At this time, the utterance processor **302** outputs the start signal to the time measurer **303** and the communicator **304**. The utterance processor **302** further outputs the start command to the communicator **304**.

[0087] The communicator **304** uses an input of the start signal from the utterance processor **302** as a trigger to change the Connection Interval for BLE. Furthermore, the communicator **304** uses the input of the start signal from the utterance processor **302** as a trigger to transmit a change instruction for the Connection Interval for BLE to the terminal device **200** (the communicator **205**) via the communication network BLE**3** (step S**502**). At this time, the change instruction is, for example, LL_CONNECTION_UPDATE_IND (a command for Link Layer). The command includes at least a value of the Connection Interval (a value within a range between 7.5 ms to 4 sec).

[0088] The terminal device **200** (the communicator **205**) receives the change instruction for the Connection Interval for BLE from the smart speaker **300** via the communication network BLE**3** (step S**503**). Then, the communicator **205** changes, after waiting for a period of time determined in the BLE standard (6 Connection Events), the value of the Connection Interval for BLE to the value included in the change instruction (steps S**504**, S**505**).

[0089] After the series of processing has been completed, BLE communication is performed between the smart speaker **300** (the communicator **304**) and the terminal device **200** (the communicator **205**) at the Connection Interval after changed.

[0090] It is assumed that, after the value of the Connection Interval has been changed, the user has uttered the music playing command toward the smart speaker **300**. Then, the microphone **301** detects the utterance of the user, and the utterance processor **302** detects the music playing command from a sound signal. At this time, the utterance processor **302** outputs the music playing command to the communicator **304**.

[0091] The communicator **304** transmits, as the music playing command is inputted from the utterance processor **302**, the inputted music playing command to the terminal device **200** at the Connection Interval after changed. The terminal device **200** generates, as the music playing command is inputted from the smart speaker **300** (the communicator **304**), music streaming data corresponding to the inputted music playing command. The terminal device **200** transmits the generated music streaming data to the smart speaker **300** (the communicator **304**) at the Connection Interval after changed.

[0092] The smart speaker **300** (the communicator **304**) outputs, as the music streaming data is received from the terminal device **200** via the communication network BLE**3**, the received music streaming data to the sound processor **306** via the arithmetic processor **305**. The sound processor **306** converts the inputted music streaming data into drive signals, and outputs the converted drive signals to the speaker **307**. The speaker **307** outputs sound on the basis of the drive signals inputted from the sound processor **306**.

[0093] Note that, in a case where the time measurer **303** has outputted the change end signal to the communicator **304**, the smart speaker **300** (the communicator **304**) uses an input of the change end signal as a trigger to change the value of the Connection Interval for BLE to the value of the Connection Interval before the change. Furthermore, the smart speaker **300** (the communicator **304**) uses the input of the change end signal as a trigger to transmit a change instruction for the Connection Interval for BLE to the terminal device **200** (the communicator **205**) via the communication network BLE**3**. At this time, the change instruction includes the value of the Connection Interval before the change.

[0094] The terminal device **200** (the communicator **205**) receives the change instruction for the Connection Interval for BLE from the smart speaker **300** via the communication network BLE**2**. Then, the communicator **205** changes, after waiting for a period of time determined in the BLE standard (6 Connection Events), the value of the Connection Interval for BLE to the value included in the change instruction.

[0095] After the series of processing has been completed, BLE communication is performed between the smart speaker **300** (the communicator **304**) and the terminal device **200** (the communicator **205**) at the Connection Interval before the change.

[Effects]

[0096] Next, effects of the communication system **2** will now be described herein.

[0097] In the present embodiment, a fact that there is a detection of an utterance in the smart speaker **300** is used as a trigger to change the Connection Interval for BLE. Thus, it is possible to perform communication between the smart speaker **300** and the terminal device **200** on the basis of a detection of an utterance at higher responsiveness. Therefore, it is possible to acquire responsiveness in accordance with an application.

[0098] In the present embodiment, a value of the Connection Interval, which is changed by the smart speaker **300** (the communicator **304**), is transmitted to the terminal device **200** (the communicator **205**) via the communication network BLE**3**. Thus, it is possible to perform communication between the smart speaker **300** and the terminal device **200** on the basis of a detection of an utterance at higher responsiveness. Therefore, it is possible to acquire responsiveness in accordance with an application.

[0099] In the present embodiment, in a case where there is no detection of an utterance in the smart speaker **300** for a predetermined period of time, the value of the Connection Interval is returned to the value before the change. Thus, it is possible to suppress excessive consumption of electric power.

[0100] Although the present disclosure has been described with reference to the embodiments, modification examples, and application examples, the present disclosure is not limited to the above-described embodiments and the like, and various modifications are possible. It should be noted that the effects described in this specification are only exemplified. Effects of the present disclosure are not limited to the effects described herein. The present disclosure may have effects other than the effects described herein.

[0101] For example, the present disclosure may also be configured as follows. [0102] (1)

[0103] A communication device including: [0104] a detector configured to detect a signal input that comes from outside; and [0105] a changer configured to use a fact that there is a detection in the detector as a trigger to change a Connection Interval for Bluetooth (registered trademark) Low Energy (BLE). [0106] (2)

[0107] The communication device according to (1), in which the detector is a sensor configured to detect a vibration applied from the outside. [0108] (3)

[0109] The communication device according to (1), in which the detector is a sensor configured to detect an utterance that has occurred at the outside. [0110] (4)

[0111] The communication device according to any one of (1) to (3), further including a transmitter configured to transmit a value of the Connection Interval, the value being changed by the changer, to an external device by using a BLE communication. [0112] (5)

[0113] The communication device according to any one of (1) to (4), in which the changer is configured to return a value of the Connection Interval to a value that is before the change is made, in a case where there is no detection of the signal input in the detector for a predetermined period of time. [0114] (6)

[0115] A communication system that includes a master device and a slave device communicable to each other by using a Bluetooth (registered trademark) Low Energy (BLE) communication, the master device including: [0116] a first detector configured to detect a signal input that comes from outside; and [0117] a first communicator configured to use a fact that there is a detection in the first detector as a trigger to transmit, to the slave device by using the BLE communication, a change instruction for a value of a Connection Interval for BLE. [0118] (7)

[0119] The communication system according to (6), in which [0120] the first communicator is configured to transmit a vibration detection information transmission request to the slave device by using the BLE communication, and receive vibration detection information as a response to the

vibration detection information transmission request from the slave device by using the BLE communication, and [0121] the master device further includes a determiner configured to perform a determination that there is an erroneous detection on a basis of a result of the detection in the first detector and the vibration detection information received from the slave device. [0122] (8)
[0123] The communication system according to (7), in which the slave device includes: [0124] a second detector configured to detect a signal input that comes from the outside; and [0125] a second communicator configured to use a fact that there is a detection in the second detector as a trigger to transmit, to the master device by using the BLE communication, a change request for the value of the Connection Interval for the BLE. [0126] (9)
[0127] The communication system according to (8), in which the second communicator is configured to receive a vibration detection information transmission request from the slave device by using the BLE communication, and transmit a result of the detection in the second detector as a response to the vibration detection information transmission request to the master device by using the BLE communication.
[0128] The present application claims the benefit of Japanese Priority Patent Application JP2022-077783 filed with the Japan Patent Office on May 10, 2022, the entire contents of which are incorporated herein by reference.
[0129] It should be understood by those skilled in the art that various modifications, combinations, sub-combinations and alterations may occur depending on design requirements and other factors insofar as they are within the scope of the appended claims or the equivalents thereof.

Claims

1. A communication device comprising: a detector configured to detect a signal input that comes from outside; and a changer configured to use a fact that there is a detection in the detector as a trigger to change a Connection Interval for Bluetooth (registered trademark) Low Energy.
2. The communication device according to claim 1, wherein the detector comprises a sensor configured to detect a vibration applied from the outside.
3. The communication device according to claim 1, wherein the detector comprises a sensor configured to detect an utterance that has occurred at the outside.
4. The communication device according to claim 1, further comprising a transmitter configured to transmit a value of the Connection Interval, the value being changed by the changer, to an external device by using a BLE communication.
5. The communication device according to claim 1, wherein the changer is configured to return a value of the Connection Interval to a value that is before the change is made, in a case where there is no detection of the signal input in the detector for a predetermined period of time.
6. A communication system that includes a master device and a slave device communicable to each other by using a Bluetooth (registered trademark) Low Energy communication, the master device comprising: a first detector configured to detect a signal input that comes from outside; and a first communicator configured to use a fact that there is a detection in the first detector as a trigger to transmit, to the slave device by using the BLE communication, a change instruction for a value of a Connection Interval for BLE.
7. The communication system according to claim 6, wherein the first communicator is configured to transmit a vibration detection information transmission request to the slave device by using the BLE communication, and receive vibration detection information as a response to the vibration detection information transmission request from the slave device by using the BLE communication, and the master device further includes a determiner configured to perform a determination that there is an erroneous detection on a basis of a result of the detection in the first detector and the vibration detection information received from the slave device.
8. The communication system according to claim 7, wherein the slave device includes: a second

detector configured to detect a signal input that comes from the outside; and a second communicator configured to use a fact that there is a detection in the second detector as a trigger to transmit, to the master device by using the BLE communication, a change request for the value of the Connection Interval for the BLE.

9. The communication system according to claim 8, wherein the second communicator is configured to receive a vibration detection information transmission request from the slave device by using the BLE communication, and transmit a result of the detection in the second detector as a response to the vibration detection information transmission request to the master device by using the BLE communication.